

QUESTION 1: 15 MARKS (START A NEW PAGE)

- a) Write in simplest form:

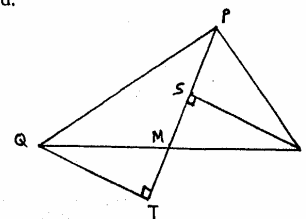
$$\frac{x+8}{x^{-1}+8^{-1}}$$
2
- b) Simplify, leaving your answer in index form:

$$\frac{25^{-1}}{(5^n)^3 \times (125)^{1-n}}$$
3
- c) Solve the following inequation and graph the solution on a number line:

$$|4-5x| < 11$$
3
- d) Find the exact length of the diagonal of a rectangle if it makes an angle of 30° with the longer side, which is 48 cm in length.
2
- e) A, B, C are the vertices of a triangle with coordinates (-2, -3), (4, 3) and (k, 5) respectively. Find:
- (i) the coordinates of the midpoint of AB, 1
 - (ii) the gradient of the interval AB, 1
 - (iii) the equation of the perpendicular bisector of AB, 1
 - (iv) hence, or otherwise, find k if $\triangle ABC$ is isosceles having base AB. 2

QUESTION 2: 15 MARKS (START A NEW PAGE)

- a) Differentiate with respect to x:
- (i) $4 \sin(3-x)$ 1
 - (ii) $\sqrt{1+x^2}$ 2
 - (iii) $3xe^{2x}$ 2
 - (iv) $\log_e(\log_e x)$ 1
- b) In $\triangle PQR$, M is the midpoint of QR, QT and RS are perpendicular lines drawn from Q and R to the line PM produced.



- (i) Prove $\triangle QMT \cong \triangle RMS$, giving reasons. 3
 - (ii) What type of quadrilateral is QSRT, giving reasons. 2
- c) For what value(s) of k will the following equation have equal roots?

$$x^2 - (k+4)x + 7 + k = 0$$
2
- d) Evaluate the following limit: $\lim_{x \rightarrow 1} \frac{x^2 - x}{1 - x}$ 2

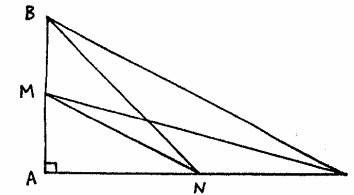
QUESTION 3: 15 MARKS (START A NEW PAGE)

- a) Determine the set of values of x for which $\frac{dy}{dx} > 0$ in the function: $y = -x^3 + 13x^2 - 35x + 5$ 3
- b) (i) Solve for x : $1 - 2\cos x = 0$, $0 \leq x \leq 360^\circ$. 2
- (ii) Sketch: $y = 1 - 2\cos x$, $0 \leq x \leq 360^\circ$. 2
- c) (i) Find the equation of the tangent to the curve $y = (2x - 1)^2$ at the point A where $x = -1$. 3
- (ii) The tangent intersects the y axis at B. Find the area of the triangle OAB, where O is the origin. 2
- d) The minute and hour hands of a clock are respectively 9cm and 6cm long. Find the distance between the ends of the hands when the time is twenty past two. (Give your answer to 1 decimal place). 3

QUESTION 4: 15 MARKS (START A NEW PAGE)

- a) In this figure, $\angle BAC = 90^\circ$, M and N are the midpoints of BA and AC respectively. Let $AM = a$ units and $AN = b$ units.

Prove: $BN^2 + CM^2 = 5 \times MN^2$



- b) A and B are the points $(-3, -1)$ and $(7, 3)$ respectively. The point P (x, y) moves so that $\angle APB = 90^\circ$. 4
- (i) Derive the equation of the locus of P. 2
- (ii) Find the centre and radius of the above circle. 3
- c) (i) A wheel has a radius of 15 cm. Through what angle (in radians) does a point on the wheel rotate if the wheel rolls 150 cm along a horizontal path? 1
- (ii) A point on the wheel was initially in contact with the ground. In rolling 150 cm show that the point on the wheel has completed more than $1\frac{1}{2}$ revolutions. 2
- (iii) What is the height above the ground of the point described in part (ii) after the wheel rolls 150 cm. (Give answer to 3 decimal places). 3

End of Exam Paper

Solutions to Yr 11 - HALF YEARLY 2004

(2U.) MATHEMATICS LEVEL

4 x 15 marks each = 60 marks total

Question 1: 15 marks

Marks

a) $\frac{x+8}{x^{-1}+8^{-1}} = \frac{x+8}{\frac{1}{x}+\frac{1}{8}} = \frac{x+8}{\frac{8x+1}{8x}} = 8x$

2

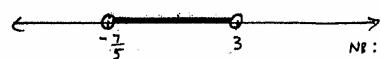
b) $\frac{25^{-1}}{(5^n)^3 \times (125)^{1-n}} = \frac{5^{-2}}{5^{3n} \times 5^{3-3n}} = 5^{-2-3n-3+3n} = 5^{-5}$

3

c) $|4-5x| < 11$
 $4-5x < 11$ OR $4-5x > -11$
 $-5x < 7$ $-5x > -15$
 $x > -\frac{7}{5}$ $x < 3$

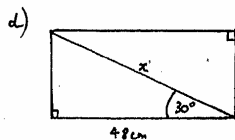
3

OR $-\frac{7}{5} < x < 3$



NB: $-\frac{1}{2}$ if closed circles

1 if graphed solution is correct from incorrect answers above
 0 if arrows are pointing wrong way.



$\frac{x}{48} = \sec 30^\circ$

1 or equivalent first step.

$x = 48 \times \sec 30^\circ$

$= 48 \times \frac{2}{\sqrt{3}}$

$= \frac{96}{\sqrt{3}}$

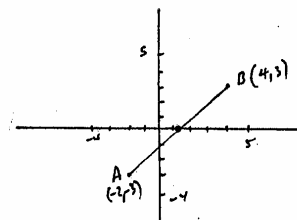
$= \frac{96\sqrt{3}}{3}$

$= 32\sqrt{3}$

1

* $-\frac{1}{2}$ if not rationalised

e)



(i) Midpoint AB = $\left(\frac{-2+4}{2}, \frac{3+3}{2}\right) = (1, 0)$

$\frac{1}{2}$ for x coord.

$\frac{1}{2}$ for y coord.

(ii) $m_{AB} = \frac{3-3}{4-(-2)} = \frac{0}{6} = 0$

(iii) $\perp m_{AB} = -1$

$\frac{1}{2}$

$\therefore y - 0 = -1(x - 1)$
 $y = -x + 1$

$\frac{1}{2}$ for any equivalent equation

(iv) The $\perp$ bisector will pass through C(k, 5), so satisfies the equation $y = -x + 1$.

1

$\therefore 5 = -k + 1$

$k = -4$

1

2

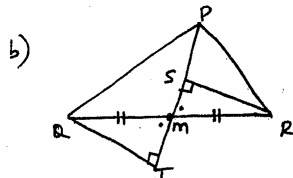
Question 2 (15 marks)

a) (i) $\frac{d}{dx} (4 \sin(3-x)) = -1 \times 4 \cos(3-x)$
 $= -4 \cos(3-x)$
 $-\frac{1}{2}$ if no negative

(ii) $\frac{d}{dx} \sqrt{1+x^2} = \frac{d}{dx} (1+x^2)^{\frac{1}{2}}$
 $= \frac{1}{2} (1+x^2)^{-\frac{1}{2}} \times 2x$
 $= x (1+x^2)^{-\frac{1}{2}}$

(iii) $\frac{d}{dx} (3x e^{2x}) = 3x \cdot 2e^{2x} + e^{2x} \cdot 3$
 $= 6x e^{2x} + 3e^{2x}$
 $= 3e^{2x} (2x + 1)$

(iv) $\frac{d}{dx} (\log_e (\log_e x)) = \frac{\frac{1}{x}}{\log_e x} = \frac{1}{x \log_e x}$
 $-\frac{1}{2}$ if left as this



(i) In Δ s QMT & SRM:
 $RM = RM$ (given) $\frac{1}{2}$
 $\angle RSM = \angle QTM = 90^\circ$ (given) $\frac{1}{2}$
 $\angle SMR = \angle QMT$ (vertically opposite) 1 *
 $\therefore \Delta QMT \equiv \Delta SRM$ (AAS) 1

* Alternatively:
 $SR \parallel QT$ (since the alternate angles RSM and MTQ are equal then the lines are parallel)
 $\therefore \angle SRM = \angle QTM$ (alternate angles equal; $SR \parallel QT$)

(ii) There are several Alternatives possible.

Solution 1:

① $SR \parallel QT$ (since the alternate angles RSM and QTM are equal, then the lines SR and QT are parallel)

② $SR = QT$ (corresponding sides of congruent Δ s)
 $\therefore$ QSRM is a parm (one pair of opposite sides equal and parallel)

Solution 2:

$SM = MT$ (corresponding sides of congruent Δ s)
 $QM = MT$ (given)

$\therefore$ QSRM is a parm (diagonals ST and QR of a parm bisect each other)

c) For equal roots, $\Delta = 0$
 $\text{ie } (k+4)^2 - 4(7+k) = 0$ ①
 $k^2 + 8k + 16 - 28 - 4k = 0$
 $k^2 + 4k - 12 = 0$ ②
 $(k+6)(k-2) = 0$ ③
 $k = -6 \text{ or } 2$

d) $\lim_{x \rightarrow 1} \frac{x(x-1)}{1-x}$
 $= \lim_{x \rightarrow 1} -x$ ①
 $= -1$ ②

Question 3 (15 marks)

a) $y = -x^3 + 13x^2 - 35x + 5$
 $\frac{dy}{dx} = -3x^2 + 26x - 35$

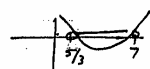
$-3x^2 + 26x - 35 > 0$

$3x^2 - 26x + 35 < 0$

$(3x-5)(x-7) < 0$

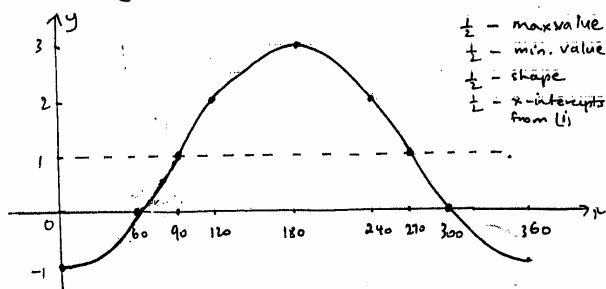
$\frac{5}{3} < x < 7$

OR $x > \frac{5}{3}$ and $x < 7$



b) (i) $1 - 2\cos x = 0$, $0 \leq x \leq 360^\circ$
 $\cos x = \frac{1}{2}$
 $x = 60^\circ, 300^\circ$

(ii) $y = 1 - 2\cos x$, $0 \leq x \leq 360^\circ$

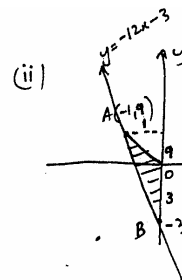


c) (i) $y = (2x-1)^2$
 $\frac{dy}{dx} = 2(2x-1) \cdot 2 = 4(2x-1)$

at $x = -1$, $\frac{dy}{dx} = 4(2(-1)-1) = -12$

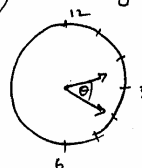
at $x = -1$, $y = (2(-1)-1)^2 = 9$

$\therefore$ Eqⁿ of tangent: $y - 9 = -12(x + 1)$
 $y = -12x - 3$



Area = $\frac{1}{2} \times 12 \times 1 = 6$ units²

d) Need angle between the hands at 2:20

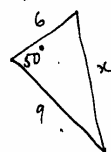


$\frac{360}{12} = 30^\circ$ for each 5 minutes.

$\frac{4}{12} \times 30^\circ = 10^\circ$

$\therefore \theta = 30 + 30 - 10$

$\theta = 50^\circ$



$x^2 = 6^2 + 9^2 - 2 \cdot 6 \cdot 9 \cdot \cos 50^\circ$

$= 47.57893815$

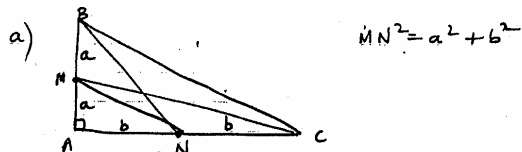
$x = 6.897748774$

$x = 6.9$ (1dp)

$\frac{1}{2}$ if not correct to 1 dp

* If angle of 50° is incorrect, max marks possible is $\frac{2}{3}$.

Question 4 (15 marks)



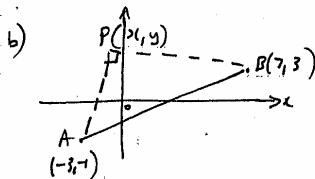
$$MN^2 = a^2 + b^2$$

In $\triangle AMN$: $a^2 + b^2 = MN^2$ ①

In $\triangle BAN$: $(2a)^2 + b^2 = BN^2$
 $4a^2 + b^2 = BN^2$ ②

In $\triangle CAM$: $a^2 + (2b)^2 = CM^2$
 $a^2 + 4b^2 = CM^2$ ③

$$\begin{aligned} \therefore BN^2 + CM^2 &= 4a^2 + b^2 + 4b^2 + a^2 \\ &= 5a^2 + 5b^2 \\ &= 5(a^2 + b^2) \\ &= 5 \times MN^2 \end{aligned} \quad \text{④}$$



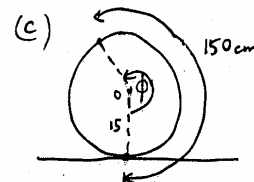
(i) Condition required: $m_{PA} \times m_{PB} = -1$

$$\frac{y+1}{x+3} \times \frac{y-3}{x-7} = -1 \quad \text{①}$$

* Any of these are possible for ① mark.

$$\begin{cases} \frac{y^2 - 2y - 3}{x^2 - 4x - 21} = -1 \\ y^2 - 2y - 3 = -x^2 + 4x + 21 \\ y^2 - 2y + 1 + x^2 - 4x + 4 = 21 + 3 + 1 + 4 \\ (y-1)^2 + (x-2)^2 = 29 \end{cases}$$

(ii) $(x-2)^2 + (y-1)^2 = 29$ ①
 Centre of circle is (2, 1), radius is $\sqrt{29}$ ①



(i) $l = r\theta$
 $150 = 15 \times \theta$
 $\theta = 10 \text{ rads}$

① if wrong units used

$\therefore$ Wheel rotates angle of 10 rads

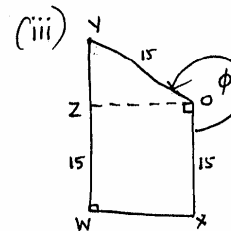
(ii) $2\pi \text{ rads} = 1 \text{ revolution}$

$1 \text{ rad} = \frac{1}{2\pi} \text{ revolutions}$

$\therefore 10 \text{ rads} = \frac{10}{2\pi} \text{ revolution}$ ①

$\therefore 10 \text{ rads} = 1.59 \text{ revolutions}$

Thus, the point has made 1.59 revolutions which is more than 1.5 revolutions. ①



$\phi = (10 - 2\pi) \text{ rads}$ ①
 (≈ 3.716814693)

$\angle YOZ = 2\pi - \phi - \frac{\pi}{2}$
 $= 2\pi - (10 - 2\pi) - \frac{\pi}{2}$
 $= 4\pi - 10 - \frac{\pi}{2}$
 $\therefore \angle YOZ = (\frac{7\pi}{2} - 10) \text{ rads}$ ①

$\frac{YZ}{15} = \sin \angle YOZ$

$\therefore YZ = 15 \sin(\frac{7\pi}{2} - 10)$

$= 12.58607294$ ①/2

$\therefore YW = 15 + 12.58607294$

$= 27.586 \text{ (3dp)}$ ①/2

$\therefore$ The height of point above ground is 27.586 cm